

Networked & Distributed Control: Assignment 2

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Abstract— This report explores various control strategies for a networked control system (NCS) with sampling irregularities, stochastic controller selection, and network-induced delays. Three main scenarios are analyzed: a periodically switching controller, a Markovian jump linear system (MJLS), and a system under delay modeled via polytopic approximation. Each scenario is modeled using appropriate state-space representations and analyzed for stability using linear matrix inequalities (LMIs). Simulations and stability regions are presented to evaluate the robustness and performance of each setup. The impact of different design choices—such as controller selection probability and delay bounds—is discussed in relation to system stability.

I. INTRODUCTION

Networked control systems (NCS) are becoming increasingly relevant in distributed and cyber-physical systems, where sensors, controllers, and actuators communicate over networks. Such systems face challenges like variable sampling intervals, packet drops, and stochastic delays, which can significantly affect stability and performance.

This report investigates these aspects by modeling and analyzing three cases:

- A periodically switching controller system with deterministic sampling intervals.
- A Markov Jump Linear System (MJLS), where controller selection follows a stochastic Markov process.
- A delayed system using polytopic approximations to account for time delays in the control loop.

For each case, the system is modeled using discrete-time state-space representations. Stability is evaluated using LMI formulations, and numerical simulations are carried out to determine the set of parameters (e.g., sampling interval h and sampling delay τ) under which the closed-loop system remains stable.

II. NETWORKED CONTROL SYSTEM

A. Q1: Switching Controllers in a Periodic Sampling Scenario

A similar plant is used as in assignment 1 so the model will not be explained again, but the same extended modeling will be used

The switching behavior is described by a periodic sequence of sampling intervals: $(h_1, h_2)^\omega$. This is captured by a finite-state automaton with two states:

- **State S1** (sampling interval h_1): applies controller $\bar{K}$ or $\bar{K}_2$
- **State S2** (sampling interval h_2): applies $\bar{K}_2$ or zero input

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The automaton will always follow the order $S1 \rightarrow S2 \rightarrow S1 \rightarrow \dots$

The switched state equations are described based on the current state, since h_1 and h_2 have different values for K . So let $x_k \in \mathbb{R}^n$ be the state at time step k . The system dynamics switch as follows:

- When h_1 : $x_{k+1} = F_1 x_k = F(h_1) - G(h_1)K_i$, where $K_i \in \bar{K}, \bar{K}_2$
- When h_2 : $x_{k+1} = F_2 x_k = F(h_2) - G(h_2)K_j$, where $K_j \in \bar{K}_2, 0$

The LMI's follow the common Lyapunov approach. The functions $P^T = P > 0$ and $Q^T = Q > 0$ must have any solution for the following inequality:

$$F_i^T \cdot P \cdot F_i - P + Q < 0$$

Where F_i and G_i is the closed loop matrix for each possible sampling time h and their controllers K . This results in 4 LMI's for the switching part and the conditions for positive P and Q results in a total of 6 LMI's

By iterating over values of $h \in [0.01, 1.0]$, the system's stability was verified using the above mentioned common Lyapunov stability. The stability region was plotted in figure 1. Where for each value of h a true or false for stability was plotted.

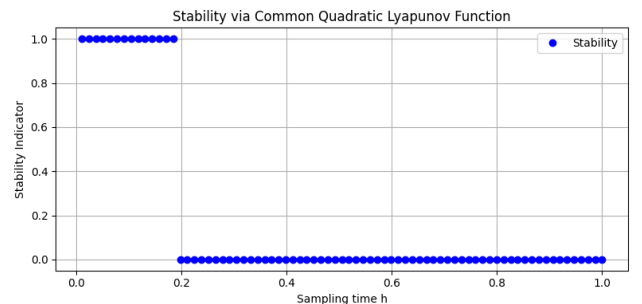


Fig. 1. The stability of the switching controller. Stability is reached at "1", while instability is at "0".

B. Q2: Markovian Switching Controller

Similarly as in section II-A switching states are used. Also the order $(h_1, h_2)^\omega$. The same states are used and defined as follows:

- S1: h_1 , controller $\bar{K}$
- S2: h_1 , controller $\bar{K}_2$
- S3: h_2 , controller $\bar{K}_2$
- S4: h_2 , zero input

But this time probabilities are used to transition between states. A picture was made to show this more clearly as a markov chain.

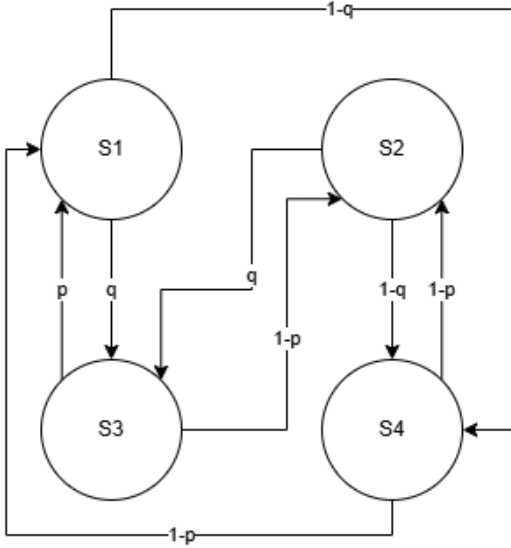


Fig. 2. The Markov chain for h1. For h2 the model will be the same but using probability q instead.

To actually use this inside the code a transition matrix will be created:

$$prob_matrix = \begin{bmatrix} 0 & 0 & q & 1-q \\ 0 & 0 & q & 1-q \\ p & 1-p & 0 & 0 \\ p & 1-p & 0 & 0 \end{bmatrix}$$

The system becomes a Markov Jump Linear System cause it uses mode-dependent linear dynamics with $A_{cl}(q_k) \in F_i - G_i K_i$, based on the current state q_k . Another requirement is a non-fixed pattern, instead using probabilities, which this system uses.

Stability is verified by use of Mean Square Stability (MSS). Since it is a MJLS, checking for MSS is verified by solving for $P_i \succ 0$ such that:

$$P_i - \sum_{j=0}^N p_{ij} A_j^T P_j A_j \succ 0$$

In code this results in:

$$P_i \cdot prob_matrix_{i,j} \cdot (A_j^T \cdot P_j \cdot A_j) \succ 0$$

Where A_j is the closed loop matrix for transition j. This is then solved for all Markov transitions j from current state i. Using CVXPY with constraints for all Markov transitions.

Now the simulation results are shown in figure 3. Where again stable is given by "True" and unstable by "False".

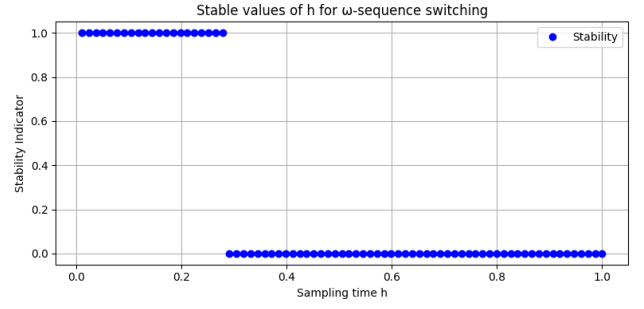


Fig. 3. The stability of the Markovian switching controller. Stability is reached at "1", while instability is at "0".

III. QUESTION 3: DELAY AND POLYTOPIC APPROXIMATION

Given new A and B , The closed loop poles will be set at $-2+j$, which means the eigenvalues must equal:

$$(s - (-2 - j))(s - (-2 + j)) = 0$$

Rewriting gives:

$$s^2 + 4s + 5 = 0$$

The eigenvalues of our actual system when using values $k1$ and $k2$ and applying $\det((A - K * B) - s * I)$ is represented by:

$$\begin{bmatrix} k1 + 1.0 * s + 2.7 & k2 \\ -1 & 1.0 * s - 1.5 \end{bmatrix}$$

This results in: $k1 = 2.8$ and $k2 = -13.25$. Sadly using this system and the existing code to plot it resulted in a system without any possibly combination of h and τ to create a stable system using the spectral radius method. This is shown in figure 4.

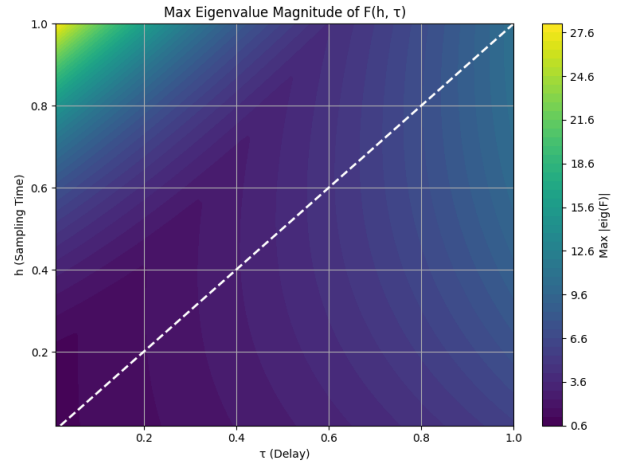


Fig. 4. The stability of the Markovian switching controller. Stability is reached at "1", while instability is at "0".

Obviously the results of this show no stable combination using the Polytopic model either since for an unstable closed loop matrix there can be no stable outcome. Instead the original system was used to plot the using the Polytopic model, resulting in figure 5 and figure 6.

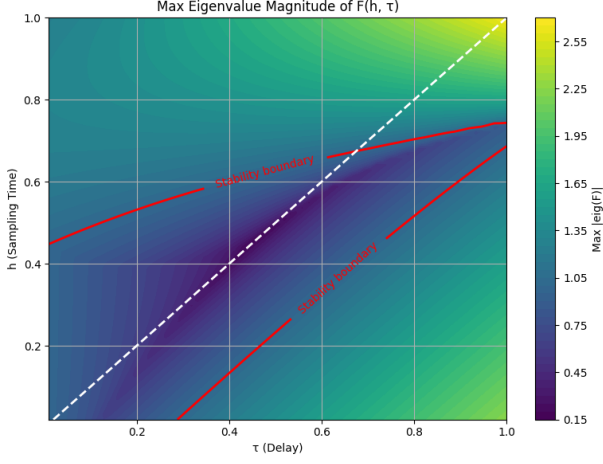


Fig. 5. The stability of the Markovian switching controller. Stability is reached at "1", while instability is at "0".

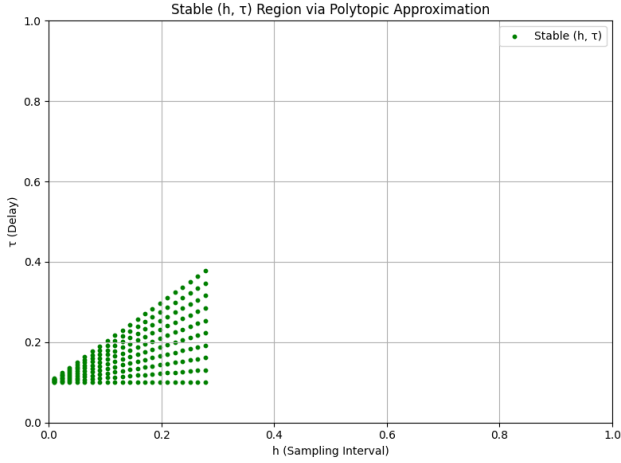


Fig. 6. The stability of the Markovian switching controller. Stability is reached at "1", while instability is at "0".

IV. DISCUSSION

The analysis of each scenario highlights different aspects of networked control systems:

- **Periodic Switching (Q1):** This deterministic setup allowed us to apply classical switched system analysis. A common Lyapunov function was sufficient for verifying stability across all controller and sampling time combinations. The LMI formulation was relatively simple and efficient to solve. For this periodic switching there is an alternative method that only uses P and no Q to reduce the amount of LMI's required even further.
- **Markovian Switching (Q2):** Introducing probabilistic switching between controllers made the system a Markov Jump Linear System. This required a more complex formulation based on Mean Square Stability (MSS). The results showed a more constrained stability region compared to the periodic case, highlighting the sensitivity of MJLS to transition probabilities p and q.
- **Polytopic Approximation under Delay (Q3):** The presence of time delays introduced additional modeling com-

plexity. Even though a controller was designed to place the poles at specific locations, the polytopic approximation revealed that the closed-loop system remained unstable across the tested delay range. This illustrates the challenge of delay compensation in NCS, even with state-feedback control.

Across all cases, simulation-based stability plots provided intuitive insights into the effect of sampling time and delays on stability. Notably, the MJLS approach showed narrower bounds for stable sampling intervals compared to the deterministic counterpart.

V. CONCLUSION

This report has presented a detailed analysis of three different control scenarios in networked systems. The periodic and Markovian switching models demonstrated the use of LMIs to verify stability under both deterministic and stochastic switching conditions. The third scenario, involving network-induced delays, showed the limitations of traditional pole placement when delays are not explicitly considered in the control design.

REFERENCES